A Multiagent-Based Methodology for Known and Novel Faults Diagnosis in Industrial Processes

Mohamed El Koujok, Ahmed Ragab, Hakim Ghezzaz, and Mouloud Amazouz

***Abstract*—This article proposes a multiagent-based methodology for the real-time fault diagnosis in industrial processes. This articles aims to build a decision support tool that helps process operators identify and better manage abnormal situations. The supervised and semisupervised machine learning methods are widely used to develop such tools. Despite their accuracy in classifying faults, supervised methods have a major limitation: they cannot diagnose novel faults. The semisupervised methods can detect and isolate novel faults but cannot disclose their root causes. The proposed methodology combines both supervised and semisupervised methods in a parallel–serial structure, exploiting their respective strengths. Moreover, it provides the process expert with the meaningful explanations of the detected novel faults or otherwise. Two case studies are used in this article to demonstrate the effectiveness of the proposed methodology. The first case is the Tennessee Eastman process benchmark. The second one uses the real data collected from a heat recovery system in a thermomechanical pulp mill.**

***Index Terms*—Artificial intelligence (AI) and machine learning (ML), augmented intelligence, decision support systems, fault detection and diagnosis (FDD), multiagent systems, process industry, pulp and paper mills, Tennessee Eastman process (TEP).**

I. Introduction

MISMANAGEMENT of abnormal/faulty situations (inefficiencies, equipment failures, and control problems) can cause significant losses to the industry. Statistics show that 70% of the industrial incidents are caused by human error [1]. One example of a very serious incident was the British Petroleum (BP) oil platform explosion, which occurred in the Gulf of Mexico in 2010. That immense disaster had a significant impact on the safety of personnel, the environment, and the economy. A BP report identified the main cause of the explosion as a pressure gauge misread by the managers, combined with a series of poor cost-cutting decisions [2]. Such disasters have highlighted the need to develop an efficient and reliable decision support tool (DST).

This need poses a major challenge for both process engineers and process operators (POs) in large-scale industrial processes, who constantly desire to be provided with pertinent and accurate information so they can maintain a high level of performance as well as better manage abnormal situations. The goal is to help them improve their diagnostic capability in real time, so they can take corrective measures while protecting safety, profitability, and environment. The researchers and practitioners in the fault detection and diagnosis (FDD) domain are constantly working on approaches that use the best available tacit and explicit knowledge. Usually, the more complex the process, the less available the tacit knowledge is. Acquiring explicit knowledge requires a deep understanding of the underlying physics of the process in order to build a rigorous mathematical model [3]. However, for a large-scale plant, constructing such models may be difficult or even impossible. Fortunately, in the context of Industry 4.0, plants are equipped with a plethora of instruments that capture hundreds of variables, hence generating huge datasets. The availability of massive amounts of data represents an important asset that could be exploited using artificial intelligence (AI) to provide POs with efficient DST for managing abnormal events. This richness of available historical data and machine learning (ML) techniques could establish the basis for the explicit knowledge required to build an efficient DST.

The supervised and semisupervised ML approaches have been widely applied for FDD [4]. The supervised approach requires labeled historical data that contain both normal operations as well as faulty situations. Once it has been trained, a supervised FDD model should be able to predict the operating conditions in real time [5]. The most common supervised methods for FDD are artificial neural networks (ANN), support vector machines (SVM), k-nearest neighbors (kNN), decision tree (DT), and the random forest (RF), all known for their effectiveness in the pattern classification [6]. The semisupervised approach requires data collected only from the normal operation condition (NOC) to build the model for abnormality detection and isolation. The most popular semisupervised FDD methods are those based on the multivariate statistical process control, such as the principal component analysis (PCA) [7]. These methods have been widely proposed by researchers and practitioners, and they were combined with contributions analysis methods (CAMs) for FDD ([8]) since they are comparatively simple to apply to real processes [9].

Manuscript received November 1, 2019; revised January 14, 2020 and March 15, 2020; accepted June 29, 2020. Date of publication August 18, 2020; date of current version February 22, 2021. This work was supported by the Program of Energy Research and Development (PERD) of Natural Resources Canada. Paper no. TII-19-4842. *(Corresponding author: Mohamed El Koujok.)*

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Digital Object Identifier 10.1109/TII.2020.3011069

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Each of these approaches has its own pros and cons. The supervised methods give a high level of precision in the diagnosis and accuracy for both normal and faulty operations; however, they may not give a detailed explanation of the obtained decisions. Moreover, they cannot diagnose the novel faults that are not present (i.e., are unknown) in the training database. In contrast, the semisupervised methods can provide interpretation and isolate novel faults.

In the literature, combining the supervised and semisupervised methods for FDD has been utilized by many researchers to exploit the advantages of the two approaches. Verron *et al.* [10] used the decomposition of Hotelling's T^2 statistic (semisupervised) and combined it with the Bayesian network classifiers (supervised) for FDD. That approach was able to detect and classify a known fault as well as identify the variables contributing to that. The approach can also detect that there is a novel fault, yet it is unable to provide information that could help diagnose it. Grbovic *et al.* [11] also proposed a method called the cold start fault detection, which combines PCA and SVM. However, that method does not provide any information about the novel faults detected.

To the best of our knowledge, the majority of these methods are limited, in which they cannot address the following needs at the same time.

1) Interpretable results that can assist POs to make a correct decision.
2) Novel fault detection and isolation.
3) High accuracy FDD in the real time.

Given these limitations, this article aims at designing and validating a methodology that integrates the supervised and semisupervised ML methods into a multiagent-based approach to diagnose the faults, whether or not they are novel. The main contribution of the proposed methodology is to connect three learning agents in a parallel-serial-layered structure for FDD; a detection agent (DTA) to detect an abnormal situation in the monitored process; a diagnosis agent (DA) that predicts not only the fault type in the real time but also provides information on the certainty of the predictions; and an isolation agent (IA) to make further explanation available to the PO by identifying the variables, which are contributing to the detected fault. This article can be considered a significant step toward the augmented intelligence in industrial processes, where the multiagent learning methods and human knowledge in the industrial process will collaborate to solve real problems. Then, the proposed methodology will allow the PO to incorporate their reason in the decision making to bring the process back to its normal operation.

In this article, the PCA is used as a semisupervised method and it is combined with a multidirectional reconstruction-based contribution (M-RBC) as CAM to isolate the faults [12], while the RF, which employs an ensemble of DT for the classification, is used as a supervised method [13]. The proposed methodology is demonstrated by using two case studies. The first case study is the Tennessee Eastman process (TEP), a well-known benchmark problem in process monitoring. The second one uses the real data in process industry: a reboiler system for heat recovery in the thermomechanical pulp (TMP) mill.

The rest of this article is organized as follows. Section II describes the different supervised and semisupervised methods. Section III explains the proposed methodology. Section IV discusses the case studies and the results obtained, and finally, Section V concludes the article.

II. Real-Time FDD in Industrial Processes

In the Industry 4.0 era, a huge amount of data can be collected and subsequently archived. The assistance of a process expert is essential to review the data and distinguish between the normal and faulty operations. The NOC dataset reflects all-natural variance for the process, while acceptable process performance is sustained. The faulty operations dataset reveal any abnormal process performance and can be categorized into several classes. It is commonly known that the semisupervised methods use only the NOC dataset to build models that can detect and isolate unacceptable process variations. On the other hand, the supervised methods are often used to classify and diagnose the faults, assuming that the types of faults have been characterized *a priori* in the database. These methods are briefly discussed in the following sections.

A. Supervised Methods for Fault Diagnosis

The supervised-learning methods are based on the analysis of relations between the predictor variables and trends in the data to predict one or more dependent variables. The ANN, SVM, *k*NN, DT, and RF are among the most popular ML methods applied for the fault diagnosis. The majority of those methods have been studied by many authors in the literature [6], [14] and it is concluded that the RF has demonstrated its competitive advantage over the others regarding the recognition of faults. In addition to the literature findings on the accuracy, stability, and robustness of the RF, we went further to compare its performance with ANN, SVM, *k*NN, and DT on the two case studies presented in Section IV in this article (see Tables I and VII). An additional advantage of the RF classifier is its ability to give the posterior probability of each class. In the context of FDD, this could provide the PO with a quantified degree of confidence in the prediction.

Consequently, the RF is selected to be integrated into the DA (see Fig. 1) for the fault diagnosis in this study, in order to give not just a prediction (classification) but also a prediction with an associated certainty. RF has several hyperparameters that must be set before the training, such as the number of trees, number of variables to select for each split, number of branch nodes (depth), and the split criterion. Also, there is a need to determine the method that aggregates the trees constructing the RF model. The performance of the RF is boosted by using an ensemble of DTs. The concept is to generate multiple versions of DTs and then combine them in an aggregated model that gives a better generalization. The idea behind splitting the space of variables randomly is to avoid constructing weak DT classifiers, which may use some uninfluential variables. Recently, an increase in computing power has resulted in a resurgence of hyperparameter optimization approaches. The goal is to reduce human effort by

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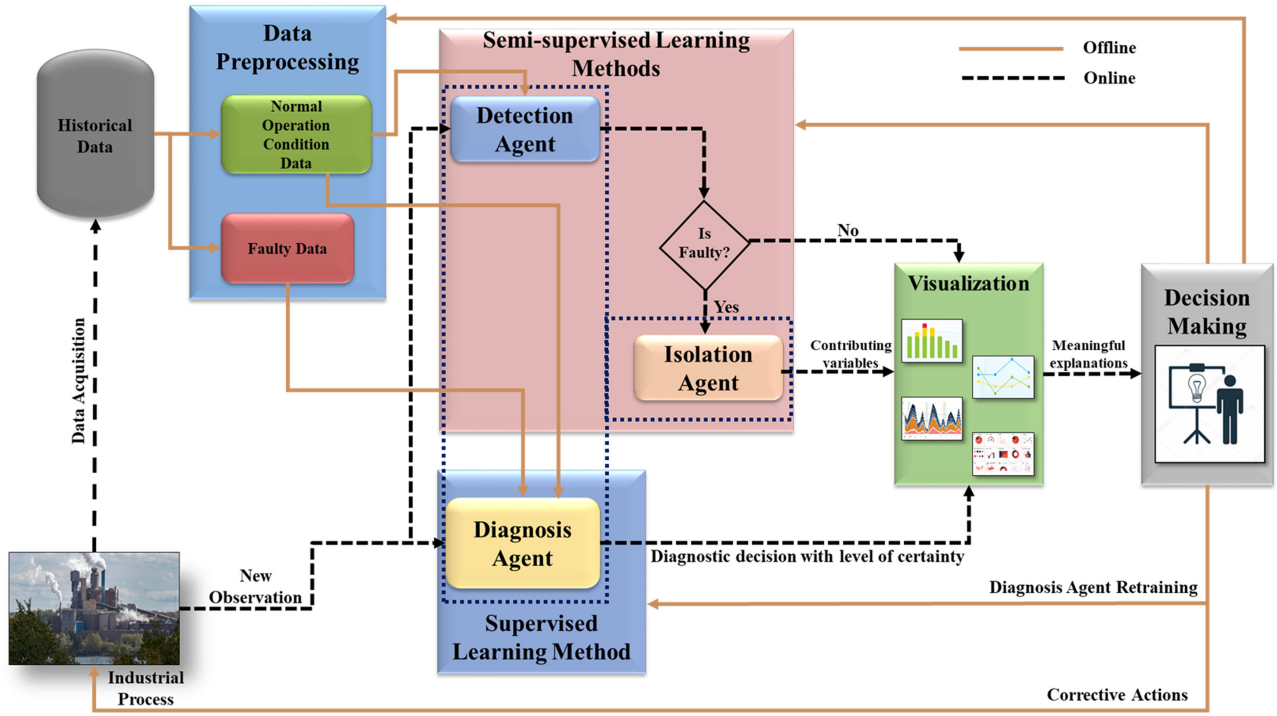

Fig. 1. Flowchart of the proposed FDD methodology.

automating the tuning of hyperparameters while avoiding overfitting or underfitting. In this article, the Bayesian optimization algorithm (BOA) is used for the optimization of hyperparameters as well as for the aggregation of trees in the RF model (for more detail about the BOA, see [15]).

B. Semisupervised Methods for Detection and Isolation

Multivariate statistical process monitoring techniques such as PCA is the most popular semisupervised method used for FDD. The PCA and its variants have been extensively used as the fault detection and isolation methods in industrial processes since they can handle the large amounts of multivariate, noisy, and incomplete historical data [7]. They transform the correlated and uncorrelated variables into a new set of latent variables, which can reflect the physical meaning in two independent spaces, each with a detection index that is associated with the fault isolation. The detection indices are used in the real time to detect deviations from the normal operation. The variables contributing to a deviation are then isolated using CAMs. The main advantage of CAMs is that they can isolate a fault without using any historical data from the faulty operation. They only require data collected from the normal operation to direct the attention of the POs during the abnormal operation to a set of faulty variables. As a result, the POs have to use their expertise to find the root cause of the faulty situation. Subsequently, the PCA was selected in this methodology to build the detection and IAs as follows.

1) Fault Detection: Two common fault detection control charts are used in the PCA: Hotelling's T^2 and the squared prediction error (SPE) [16]. Hotelling's T^2 quantifies the variance of observations captured by the principal components (PCs) subspace, while the SPE is related to the residual subspace, which measures the approximation error of the PCA model. An abnormal situation is detected when one or both indices exceeds its control limit δ. It is well known that the T^2 index is not always robust at detecting small deviations from the normal operation data, as opposed to the SPE [17]. Yue and Qin [8] proposed a new index φ, where the SPE and T^2 are combined in order to provide a complete measure of the variability in the entire space. This combined index of SPE and T^2 is deployed in the DTA.

2) Fault Isolation: Once the fault is detected, the CAMs are widely used to isolate the affected variables. These methods associate each variable with a contribution value. One of the most common CAMs methods is the contribution plot found in [18]; however, this method suffers from a smearing effect, in a way that a nonfaulty variable can have a large contribution value [19]. To resolve this problem, Alcala and Qin [20] proposed a method called the reconstruction-based contribution (RBC). It is proven in [21] that the RBC can successfully identify the faulty variable. Based on the RBC, Li *et al.* [12] proposed an improved RBC called M-RBC. This extended method for fault isolation aims to reduce the number of faulty candidates and pinpoints only the faulty variables that can help diagnose the abnormal situation. In the M-RBC method, all possible combinations of the faulty variables are explored, and finally, it can locate the ensemble of the faulty variables by reconstructing and minimizing the combined index φ. The IA in this work uses the M-RBC to provide real-time fault identification.

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III. Proposed Methodology for Real-Time FDD

The motivations and main concepts are presented in Section III-A and Section III-B demonstrates the steps.

A. Motivations and Main Concepts

One of the limitations of the approaches reported in the literature is that the supervised and semisupervised methods are usually integrated into a serial structure, such that the final decision is made based only on the supervised method, which usually represents the final stage. Another limitation is the inability to detect and diagnose the novel faults when applied to real industrial processes with noisy and inconsistent data [22]. To address these two limitations, we suggest connecting three learning agents in a parallel-serial-layered structure (see Fig. 1). The proposed methodology combined the DTA and DA in a parallel connection, where a new observation will be transmitted at the same time to be the inputs of the DTA and DA. The role of the DTA and DA is to flag if there is an abnormal situation and to diagnose the root cause. In addition, the decision of the DA is linked to a level of certainty and this information will be communicated to the PO. To make the final decision, our methodology introduces an agent (IA) to provide an explanation that helps the PO to understand the reason behind the prediction of the DA. Through this proposed structure, the PO is able to apply his expertise and interact with DST during the process monitoring.

B. Steps of the Proposed Methodology

Each agent of the proposed FDD methodology is trained and built offline using labeled historical datasets (X, C), where X contains N observations. Each observation is a row vector $(x_t)\,\epsilon\mathbb{R}^m$ and C is a vector, where each element represents a class label (a type of fault) to which the observation in X it belongs. As shown in Fig. 1, the process expert defines the boundaries of the normal operation data and abnormal (faulty) data according to the knowledge about different key performance indicators in the process. Assume that C contains K classes, where one of the classes represents the NOC. Consequently, the DTA is trained based on the NOC data. It uses the index (φ) to monitor the state of the process and to detect any deviation from the normal conditions. The DA is trained based on the K classes of data, including normal and faulty situations. It uses the RF classifier to classify new observations and predict the type of fault, if any. The IA uses the M-RBC model to obtain the variables that are contributing to the faulty situation that occurred. The following pseudocode explains the procedure of the proposed multiagent methodology.

1) $x_{\text{new}}\epsilon\mathbb{R}^m$ is a new observation collected in real time.
2) DA([RF(x_{new})]) = [NOC with p_1, …, K with p_K], where p_i is the probability of belonging and $\sum_{i=1}^{K} p_i = 1$. DTA([$\varphi(x_{\text{new}})$]) = I, where I is the value of the index.
3) If $\boldsymbol{p}_1 > \boldsymbol{p_i}$, where $i \neq 1$ and $I < \delta$, the process is "in-control", return to 1.
4) If $\boldsymbol{p}_1 < \boldsymbol{p_i}$, where $i \neq 1$ and $I > \delta$, the process is "out-of-control."
   a) FIA $[(x_{new})] = \{\text{Con}(V_1), \ldots, \text{Con}(V_{m'})\}$ = fault signature (FS), where Con is the contribution of variable V_l, $l = 1, \ldots, m'$ and $m' \in \{1, \ldots, m\}$.
   b) For $n = 2$ *to K:*
      i) $p_j = \max(p_i)$, then x_{new} probably belongs to class "j" with p_j, where $j\epsilon\{n, \ldots, K\}$.
      ii) If FS corresponds to fault "j", then the PO can make a decision that the process is out-of-control and the type of fault is discovered where it is "j". Return to 1).
      iii) Else, remove class "j" and continue the loop (return to b).
   c) If FS does not correspond to any of the existing faults, then it is highly probable that the process is affected by a novel fault. At this point, the PO can use their expertise to identify the novel fault using only the FS. Once the fault is diagnosed and its type is identified, it will be archived to retrain the RF model in order to be able to detect it automatically in the future. Return to 1.
5) If $\boldsymbol{p_1} < p_i$, where $i \neq 1$ and $I < \delta$, the process can be "out-of-control."
   a) $p_j = \max(p_i)$, then x_{new} probably belongs to class "j" with the probability p_j, where $j\,\epsilon\,\{2, \ldots, K\}$. Return to 1.
6) If $\boldsymbol{p_1} > \boldsymbol{p_i}$, where $i \neq 1$ and $I > \delta$, the process can be "out-of-control."
   a) FIA $[(x_{new})] = \{\text{Con}(V_1), \ldots, \text{Con}(V_{m'})\}$ = FS. In this case, the PO must use his expertise to diagnose the fault by assuming the FS contains the most variables that contribute to the current fault. Return to 1.

IV. Cases Studies and Results

In this section, two cases are presented to validate the proposed methodology.

A. Case Study 1: TEP

Description: The TEP is an industrial process model proposed by Vogel [23] as an FDD benchmark simulator. It consists of five main units: two-phase reactor, condenser, recycle compressor, liquid–vapor separator, and product stripper (see [23, Fig. 1]). The process produces two products from five reactants. The TEP involves 12 manipulated variables (see [23, Table III]) and 41 measured variables (see [23, Table IV]).

Data Generation and Preparation: We selected the revised MATLAB simulator of TEP, proposed by Bathelt *et al.* [24]. The simulator can generate both normal and abnormal data. We selected 12 faults to investigate the efficiency of the proposed methodology (see Table I). The selected faults represent different types: step, random variation, sticking, and slow shift. For more detail about these faults (see [23, Table VIII]). It is worth mentioning that the IDV3, IDV9, and IDV15 are not included

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TABLE I
ACCURACY OF THE FIVE CLASSIFICATION METHODS ON CASE STUDY 1

Classifiers	RF	DT	kNN	SVM	NN
Accuracy (average of the F1-scores of 11 classes)	**98.32**	97.15	91.72	91.31	87.38

in the study because the controller reacts to these faults very rapidly, and returns the process to its normal condition [25]. When a fault occurs, the measured and manipulated variables of the process are affected, and the behavior of the process becomes abnormal. A total of 12 classes of data were generated with a sample time of 3 min. The first class regroups the normal operation observations and the rest contains the observations of 11 selected faults.

Fault DA: Supervised Method (RF Classifier): As aforementioned, the advantage of the proposed methodology is its ability to inform the PO about the presence of a new fault. To test our methodology, we excluded the fault IDV6 from the training set. From the viewpoint of the DA, this fault was *a priori* unknown. Thus, only the normal data and the other 11 faults were used to build the DA. To justify the choice of the RF, we compared it with the common fault classification methods (DT, *k*NN, SVM, and ANN). About 70% of the data was used to train them. The $F1$-score was used as a metric to evaluate these five candidates. The results given in Table I show the ability of the RF to outperform the other methods. Accordingly, it was selected in this article to be integrated into the DA. It is important to mention that the BOA of MATLAB ML toolbox was used as an optimization procedure to aggregate the ensemble of trees and to obtain the optimal hyperparameters of the RF. For more detail on this method, see [13] and [26].

Fault Detection and IAs: Semisupervised Methods (PCA and M-RBC). The PCA model was built using 70% of the NOC dataset. We used the simple cross-validation technique found in [27] to determine the optimal number of the principal components that explain 80% of the variability. The index φ and its control limit δ chart were obtained, and the model that calculates the M-RBC was built.

Results, Demonstration, and Discussion: About 30% of the data was used to test the performance of trained agents. A summary of the results is given in Table II, for each fault under the column titled "*Types of fault*," 120 faulty samples were selected to test the proposed methodology. First, we compared the performance of each of the RF and PCA models, when used separately from the proposed methodology. It is given in the table that the false alarm rate is reduced to zero when the proposed methodology is applied. In addition, the proposed methodology shows increasing fault detection rates for some faults, which makes it at least as effective as the other methods (RF or PCA). Second, due to space limitations, we took the highest probability given by the DA for each sample, and then we obtained 120 values where each value is a probability. We determined the average of 120 probability values and it is listed under the column titled "*Belonging probability given by DA*" (see Table II). In addition, for each individual sample of the 120 faulty samples, the IA provided an ensemble of faulty variables, and then, 120 ensembles were obtained for each fault in Table II. Subsequently, under the column titled "*Variables contributing most to the fault*" (see Table II), we adopted that, for each fault, we kept only the variables that occurred more than 60× in 120 ensembles. Proposing multiagents (DTA, DA, and IA) in a parallel-serial structure has demonstrated better accuracy in fault detection as well as providing an interpretable ML decision. In order to demonstrate how the proposed approach exploits the

TABLE II
DETECTION AND DIAGNOSIS PERFORMANCE COMPARISON-CASE STUDY 1

Test datasets : 120 samples	**RF**	**PCA**	**Proposed methodology**		
			(DTA&DA)		**IA**
	False alarm rate			**Belonging probability given by DA**	**Variables contributing most to the fault**
NOC	0.0417	0.0590	**0**		
Types of Fault	**Fault detection rate**				
IDV1	1	0.99	**1**	0.97	F1, XC3, XMV3, XMV4, XMV7
IDV2	0.99	0.99	0.99	0.96	F2, XB1, XF1, XB2, XF2, XMV6, XMV7, XMV8
IDV4	1	1	1	0.97	T7
IDV5	1	1	1	0.91	T8
IDV7	1	1	1	0.96	XMV4
IDV8	0.975	0.99	**1**	0.85	F1, T3, F3, J1, XC1, XC2, XMV3, XMV6, XMV7, XMV8
IDV10	0.98	1	**1**	0.92	T6
IDV11	0.97	0.98	**0.98**	0.83	T7
IDV12	0.98	0.98	0.98	0.80	T8
IDV13	0.97	0.95	**0.98**	0.89	P1 , F2, T3, P2, L1, P3, J1, T4 XF1, XB2, XF2, XMV6, XMV7, XMV8
IDV14	0.99	0.99	0.99	0.84	T2, T4, F4, XMV10
IDV6 novel	-----	1	**1**	0.28 (IDV8)	F1, XMV3, F3, J1, T4, T5,
	0.90	0.98	**0.99**		

F1: A feed (stream 1), XAC: Composition of A and C feed (stream 4), P1: Reactor pressure, T2: Reactor temperature, F2: Purge rate (stream 9), T3: Separator temperature, P2: Separator pressure, L1: Stripper level, P3: Stripper pressure, F3: Stripper underflow (stream 11), J1: Compressor work, T4: Reactor cooling water outlet tempeature, T5: Condenser cooling water outlet temperature, XB1: Component B to Reactor (stream 6), XC1: Component C to reactor (stream 6), XF1: Component F to reactor (stream 6), XB2: Component B in purge (stream 9), XC2: Component C in Purge (stream 9), XF2: Component F in Purge (stream 9), T6: Temperature A and C feed (stream 4), T7: Reactor cooling water inlet temperature, F4: Reactor cooling water flow, T8: Condenser cooling water inlet temperature, XC3: Concentration of A in stream 4, XMV3: Valve position feed flow component A (stream 1), XMV4: Valve position feed flow A and C total (stream 4), XMV6: Valve position purge (stream 9), XMV7: Valve position under flow stripper (stream11), XMV8: Valve position cooling water outlet of reactor, XMV10: Valve position cooling water outlet of condenser.

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TABLE III
PROBABILITIES CALCULATED BY THE DA FOR EXAMPLE #1

Class	NOC	Faults										
		IDV1	IDV2	IDV4	IDV5	IDV7	IDV8	IDV10	IDV11	IDV12	IDV13	IDV14
Belonging Probability Given by DA	**0.87**	0	0	0	0	0	0.01	0.01	0	0.03	0.01	0.01

TABLE IV
PROBABILITIES CALCULATED BY THE DA FOR EXAMPLE #2

Class	NOC	Faults										
		IDV1	IDV2	IDV4	IDV5	IDV7	IDV8	IDV10	IDV11	IDV12	IDV13	IDV14
Belonging Probability Given by DA	0	0	0	**0.97**	0	0	0	0	0.024	0	0	0.006

TABLE V
PROBABILITIES CALCULATED BY THE DA FOR EXAMPLE #3

Class	NOC	Faults										
		IDV1	IDV2	IDV4	IDV5	IDV7	IDV8	IDV10	IDV11	IDV12	IDV13	IDV14
Belonging Probability Given by DA	**0.6**	0	0	0	0.03	0.015	0	0.11	0.023	0.18	0.015	0

advantages of various ML techniques to help the PO detect and diagnose different abnormal situations in real time, we present the following four examples.

Example #1: An observation from the normal operation was sent simultaneously to the DTA and DA. The DTA did not detect any abnormal behavior, while the DA provided the belonging probability (see Table III). The highest probability is 0.87, which confirms that the behavior of the system is normal and no action is required.

Example #2: An observation was selected from the IDV4 dataset. The DTA detected a faulty situation. As given in Table IV, the DA correctly assigned the highest probability to the fault "IDV4," which corresponds to a step change in the reactor that cooled the water inlet temperature. The DA could have misclassified this fault. For this reason, our proposed methodology presents a solution through IA, which gives a physical interpretation of the fault. In this case, the IA indicated that only the $T7$ (reactor cooling water inlet temperature) is affected (see Table II), which is the root cause. Therefore, the decision from the DA matches with FS of IDV4. Then, at this stage, the diagnosis is made and the fault IDV4 is identified as the cause of the abnormal situation. After that, an action could be taken to resolve the diagnosed fault.

Example #3: An observation was selected from the IDV10 dataset. The DTA detected this fault where Table V shows that the NOC has the highest probability and then the DA did not detect the fault. Taking into consideration only the RF as a black-box model to make decisions will lead to incorrect actions. To handle this kind of situation, the proposed methodology suggests putting the PO into the loop by providing it with the appropriate knowledge in order to make quick and accurate decisions. First, it is put forward that the DA must provide a decision and a level of certainty (see Table V). Subsequently, the decision maker can get some indication as to whether or not to trust an individual prediction from ML. As given in Table V, the probability that this fault will belong to NOC class is 0.6, which is lower than usual (see Table III). Second, the methodology integrates an IA as a technique to explain the individual prediction of the DA. As illustrated in Table II, the IA revealed that the only variable contributing to the fault is $T6$ (temperature A and C feed-stream 4). According to [23, Table VIII], the root cause of IDV10 is a random variation in the C feed temperature (stream 4). Consequently, by interpreting the knowledge provided by the DA and IA, it can be concluded that there is a fault and it is IDV10.

Example #4: For this scenario, a sample was collected from the novel fault IDV6 dataset, which does not exist in the training dataset. The root cause of this fault is a feed loss of component A in stream 1 (see [23, Table VIII]). The DTA succeeded in identifying that the selected sample is an abnormal situation. On the other hand, the DA classified the sample as IDV8 with a probability of occurring equal to 0.28 (see Table II). By using only the final decision of RF, which indicated that the fault is IDV8, it is difficult to diagnose this fault correctly. Fortunately, the proposed methodology offers the appropriate resources and information to diagnose this fault. According to Table II, the level of confidence associated with the DA decision is very low: it is just 0.28. Besides, the IA discovered that the variables that contributed most to the fault, are $F1$, XMV3, $F3$, $J1$, $T4$, and $T5$ (see Table II). The expert, in this case, must check the variation of these variables to diagnose the fault. After the presence of the IDV6 fault, the value of $F1$ decreases to zero. Consequently, by observing the $F1$, it is easy to determine the root cause of the fault, which is a feed loss of component A in stream 1.

B. Case Study 2: Heat Recovery Network in TMP

Description: In TMP, wood chips are mixed with water and sent to several cascade refiners to produce low-grade papers, such as newsprint. The TMP processes have a higher yield than the chemical pulping processes; however, they are energy

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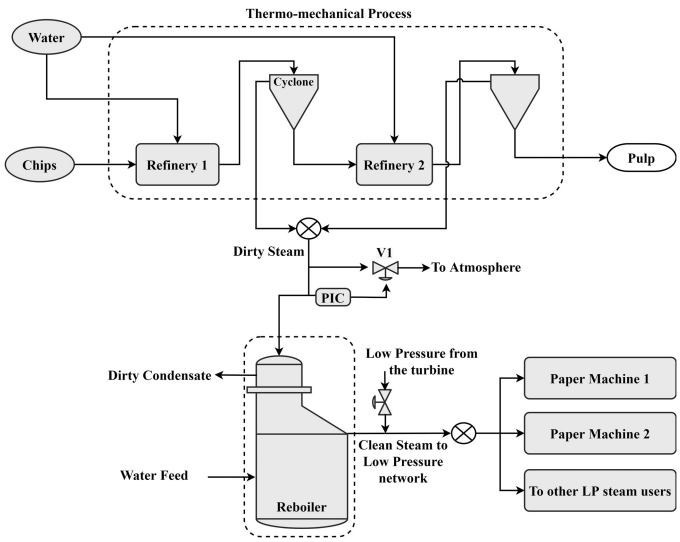

Fig. 2. Schematic view of the reboiler and its connections to the TMP process.

TABLE VI
CASE STUDY 2: NORMAL AND ABNORMAL OPERATIONS

ID	Definition	
0	Normal Operation	Vent valve is ClOSED
1	Cause 1	Paper machine 1 shut down (vent valve is OPEN)
2	Cause 2	Paper machine 2 shut down (vent valve is OPEN)

TABLE VII
ACCURACY OF THE FIVE CLASSIFICATION METHODS ON CASE STUDY 2

Classifiers	RF	DT	kNN	SVM	NN
Accuracy (Average of the F1-scores of 2 classes)	**99.65**	98.25	99.12	98.95	98.60

TABLE VIII
DETECTION AND DIAGNOSIS PERFORMANCE COMPARISON-CASE STUDY 2

Test datasets	RF	PCA	Proposed methodology		
			(DTA&DA)		IA
	False alarm rate			Belonging probability given by DA	Variables contributing most to the fault
NOC (300 samples)	0	0.09	0		
Types of Fault	Fault detection rate				
Cause 1 (100 samples)	0.99	1	**1**	0.95	P_{r1}, F_{r1}, P_{r2}, P_{r3}, P_{r4}, P_{r5}, P_{r6}
Cause 2 (100 samples)	-----	1	**1**	0.6 (Normal operation)	P_{r7}, P_{r8}, P_{r9}, P_{r10}, P_{r11}

r: Reboiler; P_{r1}: Deaerator pressure; F_{r1}: Paper machine steam flow to TMP lines; P_{r2}: Dirty steam pressure; P_{r3}: Clean steam pressure; P_{r4}: Pressure of paper machine 1 dryer section 1; P_{r5}: Pressure of paper machine 1 dryer section 2; P_{r6}: Paper machine 1 low-pressure steam consumption; P_{r7}: Low-pressure steam extraction from the turbine; P_{r8}: Pressure of paper machine 2 dryer section B; P_{r9}: Pressure of paper machine 2 dryer section A; P_{r10}: Deaerator pressure; P_{r11}: Paper machine 2 LP steam consumption (2); P_{r12}: Paper machine 2 LP steam consumption (1).

intensive. Much of this energy goes into the steam production process. Unfortunately, this steam is often contaminated with wood components and cannot be directly reused in the process. One solution is to recover this energy in the form of clean steam that could be reused to increase the overall efficiency. Energy from the dirty steam can be recovered in the reboiler system, as shown in Fig. 2. The reboiler is an indirect heat exchanger that receives hot water and dirty steam. The heat exchange between these two fluids transforms the water into clean low-pressure steam, which is mixed with a low-pressure steam from the turbines to be reused in the process. During the operation of the TMP, the pressure from the dirty steam header can increase to the point that the vent valve opens and sends dirty steam to the atmosphere, thus affecting the environment. Moreover, venting dirty steam wastes energy and reduces the energy efficiency of the overall TMP. Finding the causes of vent-valve opening would help the PO react correctly, thereby avoiding the loss of this valuable energy.

Data Preparation: A total of 24 relevant and informative process variables were selected based on expert knowledge. An inhouse software (EXPLORE) was used to clean, prepare, and separate the process data into normal and abnormal operations. Guided by the process experts, the abnormal operation (vent valve being open) is attributed to two different causes (see Table VI). It is worth mentioning that, in this study, Cause 1 is considered as a known fault, while Cause 2 is considered to be novel and it is removed during the construction of the supervised ML model.

As the first case study, 70% of the data was used to build the DTA and DA. Table VII is introduced to show the advantage of selecting the RF in this case study.

Results and Demonstration: Similar to Table II, Table VIII illustrates and compares the FDD performance of RF, PCA, and the proposed methodology using the test dataset. By combining the DA with DTA in a parallel structure, all observations belonging to the NOC test dataset were correctly classified and no false alarm was detected. On the other hand and by combining the DA with DTA, the fault detection rate increases to one. As given in Table VIII, one observation was misclassified. This is considered a Type II error, where a fault that actually exists was not reported. This is an opportunity to explain how the proposed methodology deals with the situation (see Example #5). In addition, as given in Table VIII, RF cannot diagnose Cause 2 because it has never been trained to classify this situation. However, the PCA detects that there is an abnormal situation that must be identified. Example #6 will illustrate how the information from the DA, DTA, and IA can help the PO to find the root cause

Example #5: The observation, which was misclassified by the RF, was sent to DA and DTA. As given in Table IX, the probability that this observation belongs to the "NOC" class is lower than usual. This information is essential and must be accessible to the PO. In order to make an informed decision, the proposed methodology offers additional tools, as given in Table VIII. The DTA detected that there is a faulty situation and the IA determined the significant variables (P_{r1}, F_{r1}, P_{r2}, P_{r3}, P_{r4}, P_{r5}, and P_{r6}). By interpreting this group of variables, it

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TABLE IX
PROBABILITIES CALCULATED BY THE DA FOR EXAMPLE #5

Class	NOC	Cause 1
Belonging Probability Given by DA	**0.51**	0.49

TABLE X
PROBABILITIES CALCULATED BY THE DA FOR EXAMPLE #6

Class	NOC	Cause 1
Belonging Probability Given by DA	**0.6**	0.4

is easy to associate it with the FS of Cause 1 (paper machine 1 shutdown). Consequently, our proposed design for FDD seems to be efficient and gives interesting knowledge to the PO to make better decisions in a shorter amount of time and with higher confidence.

Example #6: In this scenario, a sample was selected from the test dataset of Cause 2. Since it was removed from the training dataset, it is expected that the RF model will misclassify the actual state of the process. The proposed methodology overcomes this limitation by providing an uncertainty with the decision of the supervised ML (RF). We remarked that the DA gave 0.9 as an approximative probability when the samples belonged to NOC. However, the DA estimated that the sample belongs to NOC with the probability equal to 0.6 and then it is lower than usual (see Table X). By returning the DTA, the sample was correctly detected as faulty (see Table VIII). Subsequently, it can be concluded that the process is in an abnormal state. To diagnose the cause of this faulty situation, the PO must use the IA. As given in Table VIII, the IA pinpointed that most variables contributing to the abnormal situation are P_{r7}, P_{r8}, P_{r9}, P_{r10}, and P_{r11}. The variables do not match with the FS of Cause 1. The majority of the variables show that there is a problem with paper machine 2. Finally, the cause of this abnormal condition is identified by using the proposed methodology.

C. Discussion and Future Work

Applying the proposed methodology to a real-life case study demonstrates that it could be an effective DST for POs in the process industry. It provides interpretable knowledge to the POs and keeps humans in the loop, thus decreasing the likelihood of wrong decisions, and therefore, avoiding costly and undesired events. The reader should bear in mind that the effectiveness of the proposed methodology relies on the ability of the process industry to adopt the technologies of Industry 4.0 and mainly to equip the processes with a sufficient number of sensors that provide accurate and precise measurements. Thus, the data collected from these sensors will ease the implementation of the proposed methodology to the process industry to manage known or novel abnormal situations.

Our future activities involve improving the three actual agents to be applied to a wide spectrum of industrial applications. We will develop a meta-classifier scheme that combines the RF with other powerful supervised methods. This can significantly improve the performance of DA in complex industrial systems, in which there are different sources of data (structured and unstructured) and where severe faults compete. For the DTA, we will develop and integrate into the existing PCA other promising semisupervised methods, such as the dynamic kernel PCA. This will help the DA to handle the nonlinearities and dynamic behavior, which represent the intrinsic characteristics of most industrial processes.

To alleviate the human effort when interpreting the results of IA, we will focus on integrating a causality analysis tool into our methodology, which could assist the PO in minimizing the time required to find the root cause.

V. CONCLUSION

This article presented a fault diagnosis methodology that provides POs in the process industry with descriptive and reliable decisions. The proposed methodology integrated supervised and semisupervised ML methods in a parallel–serial structure to diagnose the faults, whether novel or not. When a fault occured, its causes were automatically identified, thus allowing the PO to take appropriate action to bring the process back to its normal operation. The methodology mainly comprised of the RF technique as a supervised ML method, PCA as a semisupervised one, and M-RBC as a CAM. The RF technique was selected because of its accuracy in diagnosing the known faults and its ability to provide a decision with a high degree of certainty. The role of the PCA and M-RBC was to detect the abnormal situation and to provide an explanation concerning the actual situation. The proposed methodology was successfully demonstrated in two case studies that use the simulated and real data. The findings show that the proposed methodology enhances the fault diagnosis by identifying known and novel faults. It also acts as a helpful DST for POs.

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